EE1003 Introduction to Computer I

Programming Assignment 1: Quintuple Stimulus Vouchers (V2)

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Due Date: 2021/10/17(Sun.) 23:59

Background

The COVID-19 broke out in mid-May 2021 in Taiwan, and nationwide level 3 alarm was held for around two months. During the difficult period, the economy was serious impacted due to the restrictions of business activities. In response to the significant economic recession, government bailouts had been issued to help stimulate the economy. In September 2021, the Quintuple Stimulus Vouchers were issued with the motivation that they would not only stimulate business opportunities, but they would also encourage people to spend more time with their family and friends, and that they can also stimulate the consumption habits of people who do not usually spend time outside. To cooperate the policy, NCU bank will be one of the banks to issue Quintuple Stimulus Vouchers.

Description

In this programming assignment, you are asked to write a program about managing Quintuple Stimulus Vouchers.

We assume a person will get Quintuple Stimulus Vouchers of total value $830. The denominations of vouchers are $1, $5, $10, $50, and $100, respectively. Each person will obtain five vouchers of each denomination. The vouchers can be stored in one's digital wallet (digital vouchers) or can be issued as printed papers (printed vouchers). If a printed voucher is used during shopping, 10% off will be given by the store. Similarly, if a digital voucher is used during shopping, 20% off will be given by NCU bank. The NCU bank will provide extra bonus for its members: If you are a member of NCU bank, you will get **EXTRA 10% off** when using the digital voucher (i.e., a 20% off first when using digital voucher, and then a 10% off if you are a member of NCU bank.)

The program will operate as follows:

1. When the program starts, a welcome message is shown and ask the user to enter the good price (positive integer) he/she want to purchase.
2. Then, the program will ask the user to select whether he/she want to use digital voucher or printed voucher. If the digital voucher is selected, additional question will pop-up to ask whether he/she is a member of NCU bank.

3. The program then will calculate the exact price (i.e., the price after discount) he/she need to pay by the discounting rule mentioned above. Note that when a decimal fraction appears, it should be (unconditionally) chopped off.
4. Then the program will process the payment. The process rules are as follows: When a bigger denomination voucher is available, you need to use it. For example, if he/she need to pay $150 and he/she owns five vouchers for each denomination, the payment should be processed as **one $100 voucher** and **one $50 voucher**. Directly using three $50 vouchers is not allowed.
5. No change will be given from the store. If he/she can exactly pay the good price, just follow the rule 4 to pay for the good. If the vouchers he/she owns cannot pay the exact price due to the lack of some denominations, always pay with bigger denomination vouchers until it is enough for the good. For example, if he/she needs to pay $39 and he/she only holds vouchers one $100, one $50, and two $5, he/she need to pay one $100 voucher for the purchase. For another example, if he/she needs to pay $156 and he/she only holds vouchers one $100, two $50, two $5, he/she need to pay one $100 and two $50 for the purchase. Note that to simplify the problem, in this system we do not consider the case that happens in the real world: if he/she holds one $50 voucher and two $5 vouchers, and the good price is $7, he/she should pay two $5 vouchers instead of one $50 voucher to minimize the loss. In out system, the above example will be processed by paying one $50 voucher for the $7 good.
6. Then the program will print the remaining vouchers he/she holds, and the program will ask the user to enter the next good price he/she want to purchase aging. Note that in the next purchase, you can change your voucher type and NCU bank member status. The corresponding discount will be recalculated.
7. The program will continuous execute until a -1 is caught from the input or he/she remains no more voucher.

As mentioned above, we assume he/she holds five vouchers of each ($1, $5, $10, $50, and $100) denomination initially. To simplify the problem, in this assignment we assume a person can only use Quintuple Stimulus Vouchers for shopping. Therefore, if the total voucher value he/she holds is less than the good price **after discount**, an error message is shown and the program will ask he/she to enter the good price again.

Input

1. A positive integer [0~ 2^{31}-1] for good price or -1 to terminate the program;
2. The type of vouchers you use, “d” for digital and “p” for printed;
3. If you choose digital vouchers, you need to input whether you are a member of NCU Bank or not, “y” for yes and “n” for no.

Output

The program will show a welcome message and prompting message at the beginning. Next, it will show the prompt to let user enter the good price. After that, the program will show the prompt to let user enter the type of vouchers. If the user chooses the digital vouchers, it will show another prompt to ask the user whether he/she is a member of NCU Bank. Then, the program will show the exact price you need to pay. If the total voucher value he/she holds is less than the good price after discount, an error message "**`You don't have enough voucher to buy the good`**" is shown and the program will ask he/she to enter the good price again. Otherwise, the program will process the payment and print the usage of the vouchers. If a denomination is not used, it will not be shown on the screen. The program will then print the remaining vouchers he/she holds. If the denomination is no longer held, it will not be shown on the screen. If he/she does not hold any voucher, the program will print "**`You do not remain any voucher`**" follows by another line "**`Bye~`**" and end its operation. Otherwise, the program will show the prompting message again for entering the good price. The program will print "**`Bye~`**" if -1 is caught as the good price, and end its execution.

Sample 1

Note: *Italic* shows the input from keyboard

```
*** Welcome to Quintuple Stimulus Vouchers System ***
Please enter the good price: 255
Please enter the type of vouchers (d/p): d
Are you a member of NCU Bank (y/n): y
The price you need to pay: 183
The payment is processed as follows:
   1 $100 voucher
   1 $ 50 voucher
   3 $ 10 voucher
   3 $  1 voucher
The remaining vouchers are as follows:
   4 $100 voucher
   4 $ 50 voucher
   2 $ 10 voucher
   5 $  5 voucher
   2 $  1 voucher

Please enter the good price: 766
```

```
Please enter the type of vouchers (d/p): p
The price you need to pay: 689
You don't have enough voucher to buy the good

Please enter the good price: 640
Please enter the type of vouchers (d/p): p
The price you need to pay: 576
The payment is processed as follows:
   4 $100 voucher
   3 $ 50 voucher
   2 $ 10 voucher
   1 $  5 voucher
   1 $  1 voucher
The remaining vouchers are as follows:
   1 $ 50 voucher
   4 $  5 voucher
   1 $  1 voucher

Please enter the good price: 40
Please enter the type of vouchers (d/p): d
Are you a member of NCU Bank (y/n): n
The price you need to pay: 32
The payment is processed as follows:
   1 $ 50 voucher
The remaining vouchers are as follows:
   4 $  5 voucher
   1 $  1 voucher

Please enter the good price: -1
Bye~
```

Sample 2

Note: *Italic* shows the input from keyboard

```
*** Welcome to Quintuple Stimulus Vouchers System ***
Please enter the good price: 1000
Please enter the type of vouchers (d/p): p
The price you need to pay: 900
```

```
You don't have enough voucher to buy the good

Please enter the good price: 789
Please enter the type of vouchers (d/p): d
Are you a member of NCU Bank (y/n): y
The price you need to pay: 568
The payment is processed as follows:
   5 $100 voucher
   1 $ 50 voucher
   1 $ 10 voucher
   1 $  5 voucher
   3 $  1 voucher
The remaining vouchers are as follows:
   4 $ 50 voucher
   4 $ 10 voucher
   4 $  5 voucher
   2 $  1 voucher

Please enter the good price: 766
Please enter the type of vouchers (d/p): p
The price you need to pay: 689
You don't have enough voucher to buy the good

Please enter the good price: 289
Please enter the type of vouchers (d/p): p
The price you need to pay: 260
The payment is processed as follows:
   4 $ 50 voucher
   4 $ 10 voucher
   4 $  5 voucher
The remaining vouchers are as follows:
   2 $  1 voucher

Please enter the good price: 3
Please enter the type of vouchers (d/p): d
Are you a member of NCU Bank (y/n): n
The price you need to pay: 2
The payment is processed as follows:
```

```
    2 $ 1 voucher
You do not remain any voucher
Bye~
```

Bonus

Anything which you think is not described within the problem specification but should be implemented in your program. For example, can your program handle the out-of-range or wrong inputs? Can your program pay the price as real world does to minimize the loss? If you add some extra function(s) in your program as a bonus, please clearly write it(them) in your report and mention to TAs during the demo session. TAs will judge the amount of bonus you get (at most 10% of this assignment) based on the difficulty and completeness of your extra function(s).

Submission Requirement

You have to submit **a source code file** (not the entire project) named as *StudID*_PA1.cpp (ex: 9862534_PA1.cpp) and **a report** named *StudID_Name*_PA1_report.pdf (ex: 9862534_ 陳 聿 廣 _PA1_report.pdf). If you implement a bonus version, please separate it from the original version. Name your source code file of the bonus version as *StudID*_PA1_bonus.cpp (ex: 9862534_PA1_bonus.cpp) and upload to ee-class with the original version. Note that the only acceptable report file format is .pdf, no .doc/.docx or other files are acceptable. **BE SURE to follow the naming rule mentioned above. Otherwise, your program will be not graded.**

We don't restrict the report format and length. In your report, you have to at least describe:

1. How to compile and execute your program; (You can use screenshot to explain)
2. The completion of the assignment; (If you complete all requirements, just specify all)
3. The hardness of this assignment and how you overcome it;
4. Bonus function(s) you implement if any
5. Any suggestions about this programming assignment?

You can put anything related to the PA in your report, such as pseudocode, control flow diagram, etc.

Your program will be judged with Code::Blocks 20.03 and GNU GCC Compiler (MinGW-W64 project version 8.1.0, 32/64 bit, SHE). We will use C++ 11 standard.

Grading

The grading is as follows:

(1) Correctness of your code: 50%
(2) Readability of your code: 10%
(3) The report: 10%
(4) Demo session: 30%
(5) Bonus (at most): 10%

Please submit your assignment on time. Otherwise, the penalty rule will apply:

- Within 72hrs delay: 20% off
- More than 3 days: 0 point

Be sure to attend a demo session (the time will be announced later). If you have questions, please E-mail to both me (andyygchen@ee.ncu.edu.tw) and TA 王啟旭 (wch861997@gmail.com)

Reference

[1] Triple Stimulus Voucher,
[on-line] https://en.wikipedia.org/wiki/Triple_Stimulus_Voucher

[2] Chapter 2 ~ Chapter 4, Paul Deitel and Harvey Deitel, "C++ how to program late objects version," 7th edition, Person 2011.